Machine Learning - Practice (HR)

Super AI Engineer Season 6

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Presented by

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DATA SCIENCE FOR BUSINESS

- Hiring and retaining employees are extremely complex tasks that require capital, time and skills.

- “Small business owners spend 40% of their working hours on tasks that do not generate any income such as hiring”.

- “Companies spend 15% -20% of the employee's salary to recruit a new candidate”.

Source: https://toggl.com/blog/cost-of-hiring-an-employee

- “An average company loses anywhere between 1% and 2.5% of their total revenue on the time it takes to bring a new hire up to speed”.

- Hiring a new employee costs an average of $7645 (0-500 corporation).

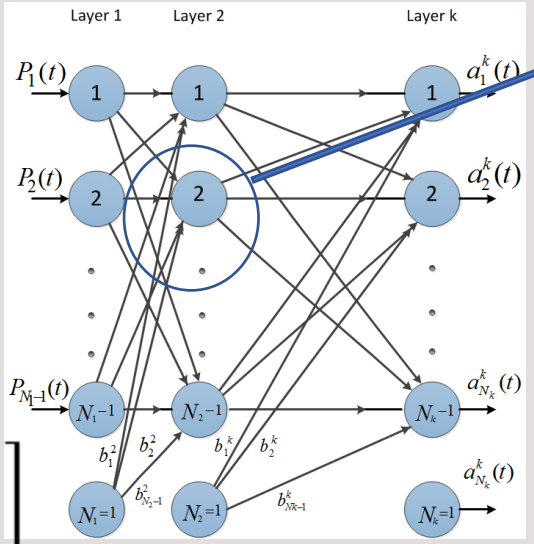
- It takes 52 days on average to fill a position.

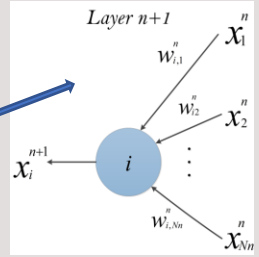
Source: https://toggl.com/blog/cost-of-hiring-an-employee

- You work as a data scientist at a multinational corporation.
- The HR team collected extensive data on their employees and approached you to develop a model that could predict which employees are more likely to quit.
- The team provided you with an extensive data, here's a sample of the dataset:
  - JobInvolvement
  - Education
  - JobSatisfaction
  - PerformanceRating
  - RelationshipSatisfaction
  - WorkLifeBalance

Data Source: https://www.kaggle.com/pavansubhasht/ibm-hr-analytics-attrition-dataset

1. ARTIFICIAL NEURAL NETWORKS





$$x_i^{n+1}(t) = \varphi\left(\sum_{j=1}^{N_n} w_{i,j}^n \, x_j^n(t)\right)$$

$$\begin{bmatrix} W_{11} & W_{12} & \cdots & W_{1,N_1} \\ W_{21} & W_{22} & & W_{2,N_1} \\ \vdots & & \ddots & \vdots \\ W_{m-1,1} & W_{m-1,2} & \cdots & W_{m-1,N_1} \\ W_{m,1} & W_{m,2} & & W_{m,N_1} \end{bmatrix}$$

Non-Linear Sigmoid Activation function

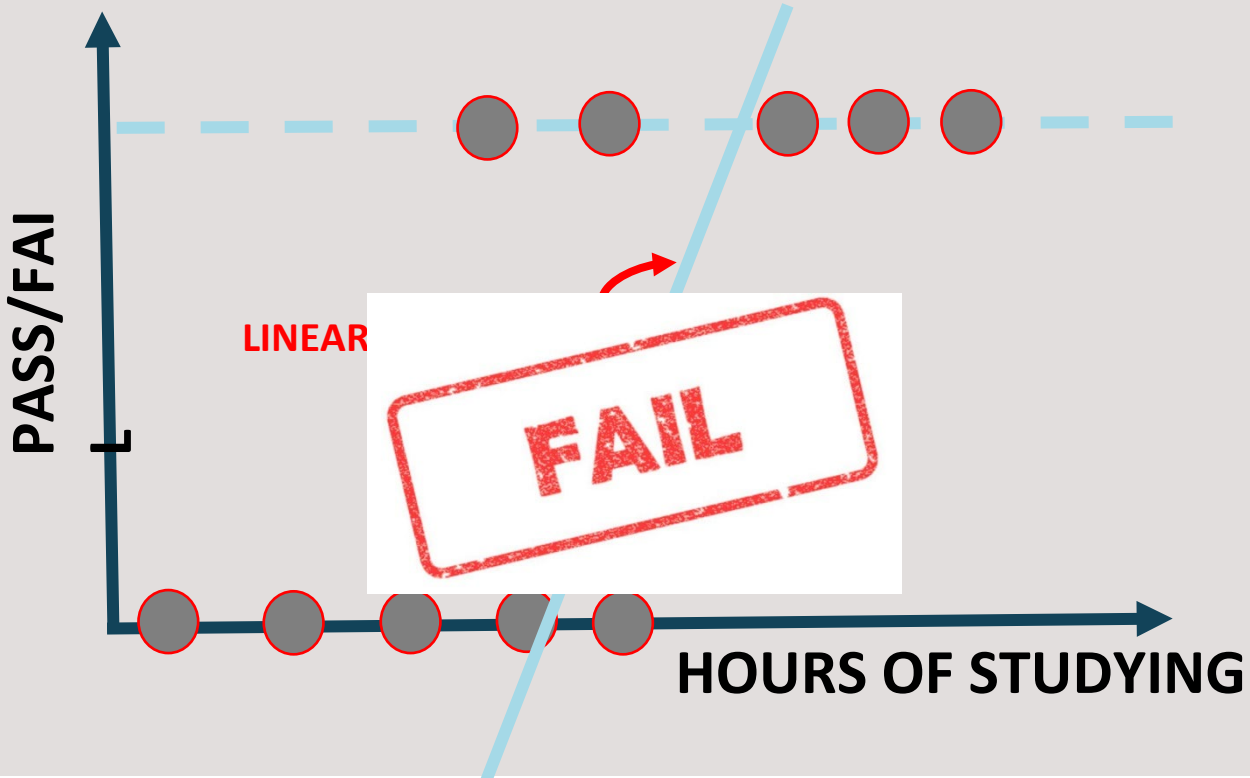
$$\varphi(w) = \frac{1}{1 + e^{-w}}$$

m: number of neurons in the hidden layer

N_1*: number of inputs*

2. LOGISTIC REGRESSION CLASSIFIER

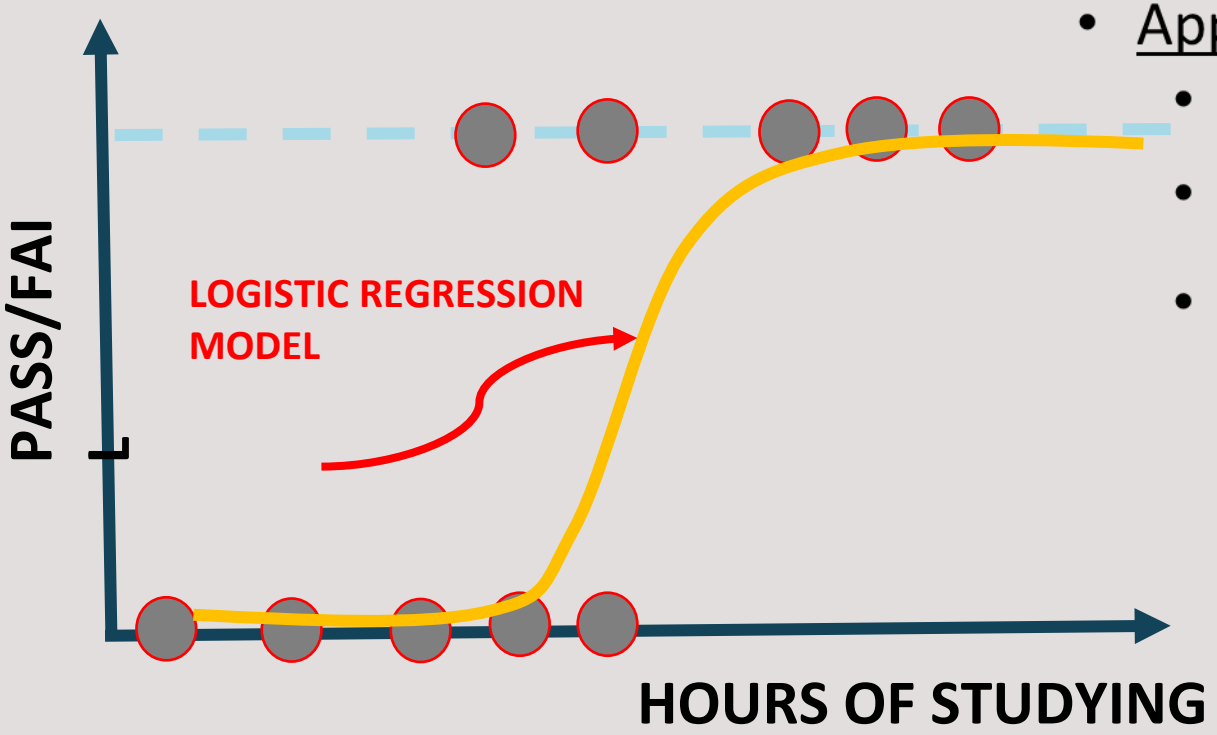
- **Linear regression** is used to predict outputs on a continuous spectrum.
  - Example: predicting revenue based on the outside air temperature.
- **Logistic regression is used to predict binary outputs** with two possible values labeled "0" or "1"
  - Logistic model output can be one of two classes: pass/fail, win/lose, healthy/sick



Hours Studying	Pass/Fail
1	0
1.5	0
2	0
3	1
3.25	0
4	1
5	1
6	1

2. LOGISTIC REGRESSION CLASSIFIER

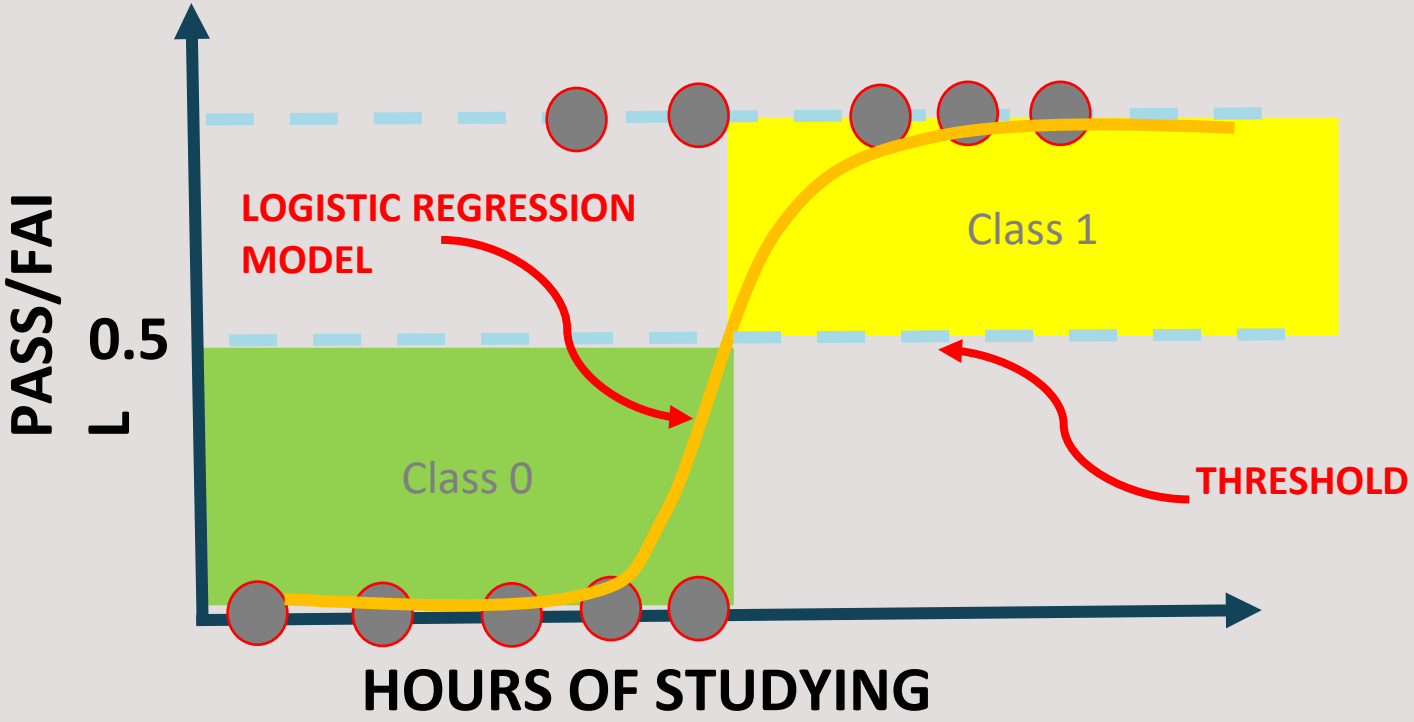
- Logistic regression algorithm works by implementing a linear equation first with independent predictors to predict a value.
- We then need to convert this value into a probability that could range from 0 to 1.



- <u>Linear equation:</u>
  - $y = b_0 + b_1 * x$
- <u>Apply Sigmoid function:</u>
  - $P(x) = sigmoid\ (y)$
  - $P(x) = \frac{1}{1+e^{-y}}$
  - $P(x) = \frac{1}{1+e^{-(b_0+b_1*x)}}$

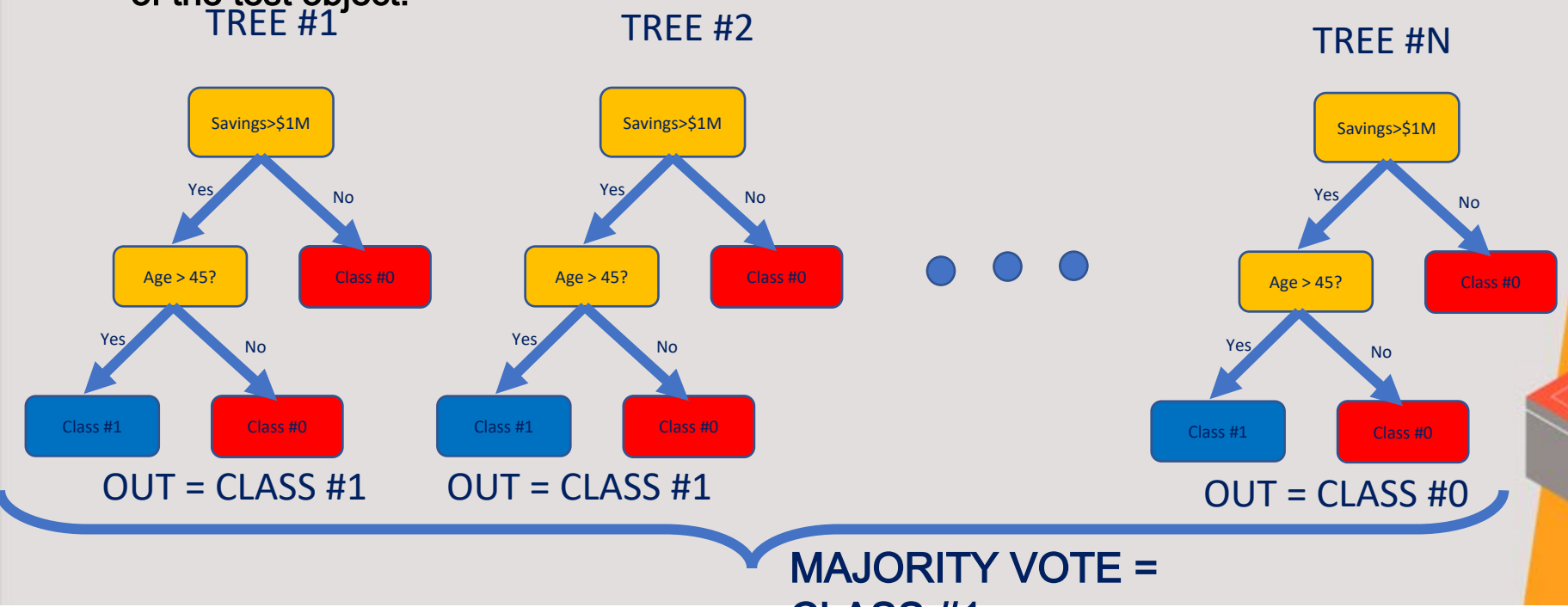
2. LOGISTIC REGRESSION CLASSIFIER

- Now we need to convert from a probability to a class value which is “0” or “1”.



3. RANDOM FOREST CLASSIFIERS

- Decision Trees are supervised Machine Learning technique where the data is split according to a certain condition/parameter.
- Let's assume we want to classify whether a customer could retire or not based on their savings and age.
- Random Forest Classifier is a type of ensemble algorithm.
- It creates a set of decision trees from randomly selected subset of training set.
- It then combines votes from different decision trees to decide the final class of the test object.



CONFUSION MATRIX

TRUE CLASS

PREDICTIONS	+	-
+	TRUE +	FALSE +
-	FALSE-	TRUE-

TYPE I ERROR

TYPE II ERROR

CLASSIFICATION MODEL KPIs

- A confusion matrix is used to describe the performance of a classification model:

  - **True positives (TP):** cases when classifier predicted TRUE (they have the disease), and correct class was TRUE (patient has disease).
  - **True negatives (TN):** cases when model predicted FALSE (no disease), and correct class was FALSE (patient do not have disease).
  - **False positives (FP) (Type I error):** classifier predicted TRUE, but correct class was FALSE (patient did not have disease).
  - **False negatives (FN) (Type II error):** classifier predicted FALSE (patient do not have disease), but they actually do have the disease
  - **Classification Accuracy** = (TP+TN) / (TP + TN + FP + FN)
  - **Precision** = TP/Total TRUE Predictions = TP/(TP+FP) (When model predicted TRUE class, how often was it right?)
  - **Recall** = TP/Actual TRUE = TP/(TP+FN) (when the class was actually TRUE, how often did the classifier get it right?)

PRECISION VS. RECALL

FACTS:
100 PATIENTS TOTAL
91 PATIENTS ARE HEALTHY
9 PATIENTS HAVE CANCER

PREDICTIONS \ TRUE CLASS	+	-
+	TP = 1	FP = 1
-	FN = 8	TN = 90

- Accuracy is generally misleading and is not enough to assess the performance of a classifier.
- Recall is an important KPI in situations where:
  - Dataset is highly imbalanced; cases when you have small cancer patients compared to healthy ones.

- Classification Accuracy = (TP+TN) / (TP + TN + FP + FN) = 91%
- Precision = TP/Total TRUE Predictions = TP/ (TP+FP) = ½=50%
- Recall = TP/ Actual TRUE = TP/ (TP+FN) = 1/9 = 11%

F1-SCORE

$$F1\ Score\ = \frac{2\ *(PRECISION\ *RECALL)}{(PRECISION + RECALL)}$$

$$F1\ Score\ = \frac{2\ *TP}{2*TP + FP + FN}$$

- F1 Score is an overall measure of a model's accuracy that combines precision and recall.
- F1 score is the harmonic mean of precision and recall.
- What is the difference between F1 Score and Accuracy?
- In unbalanced datasets, if we have large number of true negatives (healthy patients), accuracy could be misleading. Therefore, F1 score might be a better KPI to use since it provides a balance between recall and precision in the presence of unbalanced datasets.